DKU RoboMaster - Position Descriptions

Mechanics

The Mechanics Division is responsible for the physical design, construction, and optimization of our robots.

What you'll do:
- Design robot structures and mechanical systems using CAD software (e.g., SolidWorks).
- Design mechanisms for chassis, gimbal and shooting mechanisms.
- Assemble, test, and improve mechanical components.
- Collaborate with other divisions to integrate hardware and electronics.

What you'll learn:
- 3D modeling and mechanical design
- Mechanical engineering principles
- 3D printing, manufacturing, and assembly

Electronic Control

The Electronic Control Division focuses on the electrical systems and embedded programming that bring our robots to life.

What you'll do:
Develop embedded control systems using STM32.
Program and adjust motors to make them work properly.
Implement robot movement, communication, and control algorithms.
(optional) Design circuits and electronic systems.

What you'll learn:
C/C++ programming and embedded development
STM32 development and debugging
Circuit design and electronics
Motor control, sensors, and communication protocols

Computer Vision

The Computer Vision Division develops the algorithms that enable our robots to detect and track opponent robot and enable automatic aiming and targeting.

What you'll do:
Develop computer vision algorithms for target detection and tracking.
Implement image processing and recognition using tools such as OpenCV and YOLO.
Explore machine learning and AI applications in robotics using the NVIDIA Jetson platform.
Work with other divisions to integrate vision systems with robot control.

What you'll learn:
Python and C++ programming
Computer vision and image processing
OpenCV, YOLO and machine learning
Target detection, tracking, and robotics algorithms

ROBOMASTER